

A Dihedral Chow Projector and the Extreme-Hodge Gate for a Hénon μ_3 Fivefold

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Abstract

Let $K = \mathbf{Q}(\rho)$, where $\rho^2 + \rho + 1 = 0$, and consider the smooth $(2, 3)$ fivefold

$$X = \left\{ \sum_{i=0}^7 x_i^3 = \sum_{i=0}^6 x_i x_{i+1} + \rho x_7 x_0 = 0 \right\} \subset \mathbf{P}_K^7$$

arising from the fourth cohomological moment of a normalized Hénon-type Euler germ. We determine its projective monomial source stabilizer: it is an order-24 group $\mathrm{Dih}(C_{12}) = C_{12} \rtimes C_2$. Composing its Reynolds graph correspondence with explicit ordinary Chow–Künneth projectors gives two K -rational Chow idempotents on the middle cohomology. A residue-corrected Cayley-ring character calculation shows that their Hodge ranks are 10 and 158, with ledgers

$$(1, 4, 4, 1) \quad \text{and} \quad (0, 79, 79, 0).$$

The first becomes of Calabi–Yau-threefold Hodge type after one Tate twist. We also prove that rank 10 is optimal among idempotents in the rational graph algebra that retain the extreme $H^{4,1}$ line: the same trivial representation occurs four times in $H^{3,2}$. This obstruction is confined to the graph algebra. We do not claim a full automorphism classification, coniveau or an abelian realization of the complement, automorphy, a functional equation, or a new analytic half-plane.

1 Introduction

The Hilbert–Pólya dynamical-structure program explored here starts from chronological Hénon-type orbit data, forms character-weighted local determinants, and asks whether the resulting normalized Euler germ admits an arithmetic or spectral reorganization. Earlier stages of the program converted its second, third, and fourth logarithmic moments into cohomological trace packets. The fourth odd packet is

$$O_4 = H^5(X)(2),$$

where X is the fivefold in the abstract. Its Hodge types before the twist are

$$(4, 1)^1 + (3, 2)^{83} + (2, 3)^{83} + (1, 4)^1.$$

The extreme pair suggested a concrete projector gate: can source-native algebraic symmetries isolate a small Calabi–Yau-type summand?

This paper gives a deliberately two-sided answer. On the positive side, the complete projective monomial source stabilizer produces a genuine Chow projector of rank 10. On the negative side, its graph algebra cannot refine the invariant block to rank 2. The two conclusions are independent of any conjectural functional equation.

The Hodge numbers and Cayley-ring model for smooth $(2, 3)$ fivefolds are standard. Favero–Iliev–Katzarkov record $h^{4,1} = 1$, $h^{3,2} = 83$ in their Section 5.1, equation (6), and give the relevant

bigraded Cayley description in Section 5.4, equation (8) [3]. Their special diagonal basis and their Griffiths-group results are not projector theorems for the present non-diagonal quadric.

Laterveer writes the ordinary ambient Chow–Künneth projectors and middle complement in the proof of Theorem 4.1 [5]. The multiplicative result there assumes even ambient projective dimension; the odd case relevant to $X \subset \mathbf{P}^7$ is not covered. We therefore prove the required ordinary Chow compositions directly. General graph-character projectors occur in Laterveer–Nagel–Peters, Section 6, equation (4) [6], but their coniveau and finite-dimensional-motive conclusions require additional hypotheses that are not established here.

Two recent works delimit rather than supply the present theorem. Ciurca–Tanimoto–Tschinkel develop equivariant intermediate-Jacobian-torsor methods for threefold rationality and linearizability [2, §§2–3 and Theorem 3.3]; our object is a fivefold and our correspondence is a Chow graph average. Xu studies natural Chow–Künneth decompositions and motivic multiplicativity defects for complete intersections [7, Theorems 1.4 and 4.19], but does not produce the source-specific dihedral middle split used here.

High symmetry can produce interesting factors in other Fano settings: Iliev–Roulleau obtain an invariant abelian ninefold from a $\mathrm{PSL}_2(\mathbf{F}_{19})$ -symmetric cubic sevenfold [4, abstract, Theorem 1, and Corollary 10]. This provides a conceptual neighbor, not a result about our $(2, 3)$ fivefold or its field of definition.

Main contributions

1. We exhaust the phase equations and prove that the projective monomial source stabilizer is the order-24 group $\mathrm{Dih}(C_{12})$.
2. We give explicit K -rational Chow idempotents π_{core} and π_{lev} , with no MCK assumption.
3. We include the determinant ratio $\det(M_g)/\det(A_g)$ required by the residue action and compute the exact $\mathrm{Dih}(C_{12})$ -character on the Cayley quotient.
4. We obtain the rank-10/rank-158 Hodge splitting and prove that rank 10 is sharp inside $\mathbf{Q}[G]$.

The term *source* is essential throughout: we classify projective monomial transformations preserving the displayed pair of equations. We do not classify the full automorphism group of X .

Claim boundary

The same K -rational Chow idempotent acts in every standard realization, and its ℓ -adic action is G_K -equivariant. We do not call this a computed strict compatible system: common local Frobenius polynomials are a successor experiment. Nor do we infer an actual Calabi–Yau threefold, an abelian 79-fold over K , coniveau, automorphy, a Hasse–Weil functional equation, a new Euler continuation domain, or a Hilbert–Pólya operator.

2 The source and the main theorem

Fix

$$K = \mathbf{Q}(\rho), \quad \rho^2 + \rho + 1 = 0,$$

and define

$$C(x) = \sum_{i=0}^7 x_i^3, \quad Q(x) = \sum_{i=0}^6 x_i x_{i+1} + \rho x_7 x_0. \tag{1}$$

Let $X = \{C = Q = 0\} \subset \mathbf{P}_K^7$. Characteristic-zero smoothness is inherited from the preceding fourth-moment construction; the release machine source lock records the exact C50 certificate

path and SHA-256. The variety has dimension 5, degree 6, and middle Hodge numbers

$$h^{4,1} = h^{1,4} = 1, \quad h^{3,2} = h^{2,3} = 83.$$

A *projective monomial source symmetry* is a projective transformation represented by

$$x_i \mapsto \rho^{e_i} x_{\sigma(i)}$$

that preserves C and Q up to their individual nonzero scalars. Let G_{mon} be the group of all such transformations.

Put $h = c_1(\mathcal{O}_X(1))$. Define

$$\pi_{2i} = \frac{1}{6} h^{5-i} \times h^i \quad (0 \leq i \leq 5), \quad \pi_5 = \Delta_X - \sum_{i=0}^5 \pi_{2i}, \quad (2)$$

and

$$e_G = \frac{1}{24} \sum_{g \in G_{\text{mon}}} [\Gamma_g], \quad \pi_{\text{core}} = \pi_5 e_G, \quad \pi_{\text{lev}} = \pi_5 - \pi_5 e_G. \quad (3)$$

Theorem 2.1 (Dihedral middle splitting). *The projective monomial source stabilizer is*

$$G_{\text{mon}} \cong \text{Dih}(C_{12}) = C_{12} \rtimes C_2, \quad |G_{\text{mon}}| = 24.$$

The Reynolds correspondence e_G is a self-transpose idempotent. The two middle correspondences π_{core} and π_{lev} are mutually orthogonal self-transpose idempotents in $\text{CH}^5(X \times_K X)_{\mathbf{Q}}$. Their middle realizations have Hodge ledgers

$$\begin{aligned} \pi_{\text{core}} H^5(X) &: (4, 1)^1 + (3, 2)^4 + (2, 3)^4 + (1, 4)^1, \\ \pi_{\text{lev}} H^5(X) &: (3, 2)^{79} + (2, 3)^{79}. \end{aligned}$$

In particular, their ranks are 10 and 158. After one Tate twist the first ledger is

$$(3, 0)^1 + (2, 1)^4 + (1, 2)^4 + (0, 3)^1.$$

The same Chow projectors split every standard Weil realization; each ℓ -adic projector is G_K -equivariant.

Theorem 2.2 (Graph-algebra optimum). *Let $q \in \mathbf{Q}[G_{\text{mon}}]$ be an idempotent whose graph correspondence acts as the identity on $H^{4,1}(X)$. Then*

$$\text{rank}(qH^5(X)) \geq 10.$$

The bound is attained by e_G . In particular, the rational graph algebra cannot isolate only the rank-two extreme Hodge pair.

Theorem 2.2 is confined to graph correspondences in $\mathbf{Q}[G_{\text{mon}}]$. It does not exclude a K -rational algebraic correspondence outside that algebra, a projector after extending coefficients, or a different categorical construction.

3 The projective monomial source group

3.1 Finite reduction

Suppose a monomial transformation $x_i \mapsto a_i x_{\sigma(i)}$ sends C to αC . Coefficient comparison gives $a_i^3 = \alpha$ for all i . After a common projective rescaling, it therefore has the form

$$x_i \mapsto \rho^{e_i} x_{\sigma(i)}, \quad e_i \in \mathbf{F}_3, \quad e_0 = 0. \quad (3.1)$$

The support of Q is an eight-cycle, so σ lies in its order-16 dihedral automorphism group. Write

$$r_k(i) = i + k, \quad s_k(i) = k - i \pmod{8}.$$

Let $c(E) \in \mathbf{F}_3$ be zero on the seven ordinary edges and one on the closing edge $\{7, 0\}$. Preservation of Q up to ρ^{qg} is equivalent to

$$c(\{i, i+1\}) + e_i + e_{i+1} = q_g + c(\{\sigma(i), \sigma(i+1)\}) \quad (4)$$

for all cyclic indices i . This is an affine linear system over $\mathbf{F}_3$.

Proposition 3.1. *The only support permutations admitting solutions to (4) are*

$$r_0, r_2, r_4, r_6, s_1, s_3, s_5, s_7.$$

Each admits exactly three normalized phase vectors. Hence $|G_{\text{mon}}| = 24$.

Proof. Row reduction of (4) for the sixteen possible support permutations yields the complete table in Appendix A. Substitution verifies every surviving row, and the exhausted support list proves completeness. $\square$

3.2 Presentation

Choose

$$\begin{aligned} r : \quad \sigma &= (6, 7, 0, 1, 2, 3, 4, 5), \quad e = (0, 1, 1, 0, 1, 0, 1, 0), \\ s : \quad \sigma &= (7, 6, 5, 4, 3, 2, 1, 0), \quad e = (0, 1, 0, 1, 0, 1, 0, 1). \end{aligned} \quad (3.2)$$

Direct substitution gives

$$C(rx) = C(sx) = C(x), \quad Q(rx) = Q(sx) = \rho Q(x).$$

Exact phase-permutation multiplication gives

$$r^{12} = s^2 = 1, \quad srs = r^{-1}. \quad (3.3)$$

The twelve r^k and twelve sr^k are distinct. By Proposition 3.1, they exhaust G_{mon} . This proves

$$G_{\text{mon}} \cong \text{Dih}(C_{12}) = C_{12} \rtimes C_2.$$

We always specify order 24, since dihedral-group notation is not uniform across the literature.

Remark 3.2. The proof uses the particular embedding of the two equations in $\mathbf{P}^7$. It does not rule out nonmonomial automorphisms of X .

4 A Chow-level middle splitting

We now prove that the projector is algebraic before taking any cohomological realization. Since $\deg X = 2 \cdot 3 = 6$,

$$\int_X h^5 = 6. \quad (4.1)$$

Lemma 4.1. *The correspondences π_{2i} in (2) are mutually orthogonal idempotents in $\text{CH}^5(X \times_K X)_{\mathbf{Q}}$.*

Proof. For decomposable correspondences on a smooth fivefold,

$$(a \times b) \circ (c \times d) = \left(\int_X da \right) c \times b. \quad (4.2)$$

Putting $a = h^{5-i}$, $b = h^i$, $c = h^{5-j}$, and $d = h^j$, the integral vanishes unless $i = j$ and equals 6 if $i = j$. The normalizing factors in (2) therefore give

$$\pi_{2i} \circ \pi_{2j} = \delta_{ij} \pi_{2i}.$$

□

It follows immediately that π_5 is an idempotent orthogonal to the six ambient projectors. Weak Lefschetz identifies the nonmiddle cohomology with the ambient powers of h , so π_5 realizes precisely the middle cohomology. Also $\pi_{2i}^t = \pi_{10-2i}$, and therefore $\pi_5^t = \pi_5$.

Lemma 4.2. *The graph average e_G commutes with π_5 , satisfies $e_G^2 = e_G = e_G^t$, and $\pi_{\text{core}}, \pi_{\text{lev}}$ in (3) are mutually orthogonal self-transpose idempotents.*

Proof. Every $g \in G_{\text{mon}}$ is induced by a projective linear transformation, so $g^*h = h$. Thus its graph commutes with each π_{2i} and with π_5 . Moreover,

$$[\Gamma_g][\Gamma_{g'}] = [\Gamma_{gg'}], \quad [\Gamma_g]^t = [\Gamma_{g^{-1}}]. \quad (4.3)$$

In the double sum defining e_G^2 , each group element occurs exactly 24 times. This proves idempotence. Inversion permutes the group, proving self-transposition. Commutation with π_5 then proves all claims by direct expansion. □

Remark 4.3. This argument proves ordinary Chow projector identities modulo rational equivalence. It does not assert compatibility with the small diagonal, and therefore does not invoke an MCK theorem. This distinction is essential because the even-ambient-dimension hypothesis in [5, Theorem 4.1] is not satisfied by $X \subset \mathbf{P}^7$.

5 The residue-twisted Cayley character

5.1 The quotient and the projective action

Set

$$\mathcal{F} = yC + zQ, \quad R = K[x_0, \dots, x_7, y, z]/J(\mathcal{F}),$$

with bidegrees

$$\deg x_i = (0, 1), \quad \deg y = (1, -3), \quad \deg z = (1, -2). \quad (5.1)$$

The Cayley description [3, §5.4, equation (8)] gives

$$H_{\text{prim}}^{4,1}(X) \cong R_{1,-3}, \quad H_{\text{prim}}^{3,2}(X) \cong R_{2,-3}. \quad (5.2)$$

Choose a lift $M_g \in \text{GL}_8(K)$ and let A_g be defined by

$$\begin{pmatrix} C \\ Q \end{pmatrix} (M_g x) = A_g \begin{pmatrix} C \\ Q \end{pmatrix} (x). \quad (5.3)$$

The induced Cayley transformation is

$$(x, (y, z)^t) \longmapsto (M_g x, A_g^{-t}(y, z)^t). \quad (5.4)$$

The residue class has the additional orientation multiplier

$$\omega(g) = \frac{\det M_g}{\det A_g}. \quad (5)$$

Lemma 5.1. *The action consisting of polynomial pullback under (5.4) and the multiplier $\omega(g)$ in (5) is independent of the scalar lift M_g .*

Proof. Replacing M_g by λM_g replaces A_g by $\text{diag}(\lambda^3, \lambda^2)A_g$. For a representative in $R_{p,-3}$, the simultaneous Cayley polynomial pullback has total scalar λ^{-3} : the x -, y -, and z -degrees combine using the second bidegree -3 . The determinant ratio changes by $\lambda^{8-3-2} = \lambda^3$, so the factors cancel. For the explicit control $\lambda = 2$, these are 2^{-3} and 2^3 . $\square$

For the chosen generators,

$$A_r = A_s = \text{diag}(1, \rho), \quad \det M_r = \det M_s = \rho,$$

so $\omega(r) = \omega(s) = 1$. This accidental simplification does not justify removing (5).

5.2 Exact character

The ambient bidegree- $(2, -3)$ monomials form

$$y^2 K[x]_3 \oplus yz K[x]_2 \oplus z^2 K[x]_1,$$

of dimension

$$\binom{10}{3} + \binom{9}{2} + 8 = 164.$$

The corresponding Jacobian relations have rank 81, hence $\dim R_{2,-3} = 83$. Exact quotient matrices give

$$\text{tr}(r^k) = (83, -1, -3, -1, -7, -1, 3, -1, -7, -1, -3, -1), \quad (5.5)$$

$$\text{tr}(sr^k) = 3 \quad (0 \leq k \leq 11). \quad (5.6)$$

The four one-dimensional characters, ordered by

$$(r, s) \mapsto (1, 1), (1, -1), (-1, 1), (-1, -1),$$

occur with multiplicities

$$(4, 1, 3, 3). \quad (5.7)$$

For $1 \leq j \leq 5$, let

$$\vartheta_j(r^k) = 2 \cos\left(\frac{\pi j k}{6}\right), \quad \vartheta_j(sr^k) = 0.$$

Their multiplicities are

$$(7, 8, 6, 8, 7). \quad (5.8)$$

The dimension check is

$$4 + 1 + 3 + 3 + 2(7 + 8 + 6 + 8 + 7) = 83.$$

Finally, $R_{1,-3} = Ky$ is the trivial representation. The group average therefore retains one dimension in $H^{4,1}$, four in $H^{3,2}$, and their conjugates. This proves the first Hodge ledger in Theorem 2.1. Subtracting from the total $(1, 83, 83, 1)$ gives the second.

Remark 5.2. The release computation must reconstruct the quotient action and the character over an exact coefficient field. A trace list supplied by the producer without an independent quotient replay is not sufficient evidence.

6 The graph-algebra optimum

Let

$$\varepsilon : \mathbf{Q}[G_{\text{mon}}] \longrightarrow \mathbf{Q}, \quad \varepsilon \left(\sum_g a_g g \right) = \sum_g a_g$$

be the augmentation. If V_{triv} is any copy of the trivial representation, then every $a \in \mathbf{Q}[G_{\text{mon}}]$ acts on it by the single scalar $\varepsilon(a)$.

Proof of Theorem 2.2. If $q^2 = q$, then

$$\varepsilon(q)^2 = \varepsilon(q),$$

so $\varepsilon(q)$ is either zero or one. The line $H^{4,1}(X)$ is trivial. Requiring q to act as the identity there forces $\varepsilon(q) = 1$. Equation (5.7) shows that $H^{3,2}(X)$ contains four more trivial copies, and q must act as the identity on all of them. Because q is a rational graph correspondence, the conjugate pieces are retained as well. Thus

$$\text{rank}(qH^5(X)) \geq 1 + 4 + 4 + 1 = 10.$$

The Reynolds idempotent e_G acts as the identity exactly on the invariant subspace, whose rank is 10, so the bound is sharp. $\square$

This obstruction is representation-theoretic, not a universal indecomposability theorem. In particular, it does not control correspondences outside the image of $\mathbf{Q}[G_{\text{mon}}]$ in $\text{CH}^5(X \times_K X)_{\mathbf{Q}}$. It also does not control projectors after a coefficient extension.

6.1 What the Hodge ledgers do not imply

After one Tate twist, the invariant core has Hodge numbers

$$h^{3,0} = h^{0,3} = 1, \quad h^{2,1} = h^{1,2} = 4.$$

We call this a Calabi–Yau-threefold Hodge type only. No threefold realizing it is constructed.

After the C51 twist by two, the complement has Hodge types

$$(1, 0)^{79} + (0, 1)^{79}.$$

Over $\mathbf{C}$, polarizable effective weight-one Hodge structures are related to abelian varieties up to isogeny, but descent of an arbitrary G_K -representation to an abelian variety over K is an extra arithmetic statement; see the discussion in [1, Introduction, pp. 2–3]. Theorem A and Theorem 6.1 of that work give sufficient constructions under additional geometric hypotheses, such as coniveau or a full level-one complete-intersection packet; they do not apply automatically to our projected summand. We have proved neither the needed geometric realization nor coniveau one. Thus the notation A_{79}/K would be unjustified here.

7 Route-A impact and the next gate

The result is structural rather than analytic. It preserves the chronological fourth-moment source and refines its odd cohomological packet by algebraic correspondences. This is a substantial improvement to the arithmetic organization of the candidate, but it does not by itself improve the normalized Euler germ.

Criterion	C52 status
A1	Source chronology inherited unchanged from C51.
A2	Analytic determinant and normalized logarithmic germ inherited; no new determinant theorem.
A3	Algebraic packet control improved by the rank-10/158 Chow splitting; no local Frobenius polynomials, automorphy, full functional equation, or new continuation domain.
A4	A4_NATURAL_QUANTIZATION is inherited because the source two-step quantization is unchanged; no self-adjoint global generator is constructed.

The overall Route-A status is therefore exploratory. There is no Riemann-divisor match: in particular, the paper makes no claim that the center or zeros of a Hasse–Weil factor, even if later constructed, already match the Riemann critical line.

7.1 C53 gate

The next paper must open a different door. It should compute full local Frobenius polynomials of the rank-10 core at good split primes and test the commutant beyond $\mathbf{Q}[G_{\text{mon}}]$. A positive theorem would construct a new K -rational algebraic correspondence refining the trivial isotypic block. A negative theorem would prove, from exact local polynomials and a controlled correspondence algebra, that no such low-rank refinement occurs there.

Neither full local polynomial data nor an incidence correspondence is claimed in C52. Consequently C52 proves no automorphy, Hasse–Weil functional equation, or new analytic half-plane.

8 Limitations

The group enumeration is complete only for projective monomial transformations preserving the frozen source equations up to scalar. The character calculation depends on exact quotient linear algebra and must remain tied to the residue determinant ratio. The optimality theorem applies only to the rational graph algebra. The Hodge ledgers do not establish coniveau, finite-dimensionality, abelian type, or geometric realization by lower-dimensional varieties.

On the dynamical side, the construction analyzes one cohomological packet extracted from the fourth logarithmic moment. It does not produce a global self-adjoint operator, a functional equation for the full Hénon object, or a proof of the Riemann hypothesis.

Data availability

The exact source equations, phase table, character ledger, and proof specification are included in the project directory. Release certificates, independent checks, mutation tests, and the final manifest are archived with the paper. The compilation and integrity reports record the immutable artifact hashes.

Ethics statement

This research is mathematical and computational. It involved no human participants, personal data, animals, or biological materials.

Author contributions

Conceptualization, formal analysis, methodology, software design, validation design, writing, and project administration were carried out within the Hilbert–Pólya Dynamical Structure Exploration Project. Human review is required before external submission.

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AI-use statement

Generative AI systems assisted with conjecture generation, exact derivation organization, software drafting, literature routing, and manuscript preparation. Mathematical claims are separated from machine evidence and are subject to exact producer/checker replay and human verification. AI assistance is not treated as evidence for a theorem.

A Exact phase and character tables

An eight-digit word records $(e_0, \dots, e_7)$.

Support permutation	Normalized phase words
r_0	00000000, 01010101, 02020202
r_2	01010110, 02020211, 00000012
r_4	01011010, 02021111, 00001212
r_6	01101010, 02111111, 00121212
s_1	01101010, 02111111, 00121212
s_3	01011010, 02021111, 00001212
s_5	01010110, 02020211, 00000012
s_7	00000000, 01010101, 02020202

For the rotation generator, the exact $R_{2,-3}$ traces are

k	0	1	2	3	4	5	6	7	8	9	10	11
$\text{tr}(r^k)$	83	-1	-3	-1	-7	-1	3	-1	-7	-1	-3	-1

and $\text{tr}(sr^k) = 3$ for all k . The resulting multiplicities are

$(\chi(r), \chi(s))$	(1, 1)	(1, -1)	(-1, 1)	(-1, -1)
multiplicity	4	1	3	3

for the one-dimensional characters and

j	1	2	3	4	5
multiplicity of ϑ_j	7	8	6	8	7

for the two-dimensional characters $\vartheta_j(r^k) = 2 \cos(\pi j k / 6)$.

References

- [1] Jeffrey D. Achter, Sebastian Casalaina-Martin, and Charles Vial. Distinguished models of intermediate Jacobians. *Journal of the Institute of Mathematics of Jussieu*, 19:891–918, 2020.
- [2] Tudor Ciurca, Sho Tanimoto, and Yuri Tschinkel. Intermediate Jacobians and linearizability. arXiv:2403.06047v2, 2024.
- [3] David Favero, Atanas Iliev, and Ludmil Katzarkov. On the Griffiths groups of Fano manifolds of Calabi–Yau Hodge type. arXiv:1212.2608, 2012.
- [4] Atanas Iliev and Xavier Roulleau. On the $\mathrm{PSL}(2,19)$ -invariant cubic sevenfold. arXiv:1301.1142, 2013.
- [5] Robert Laterveer. Algebraic cycles and intersections of a quadric and a cubic. arXiv:2105.02224, 2021.
- [6] Robert Laterveer, Jan Nagel, and Chris Peters. On complete intersections in varieties with finite-dimensional motive. arXiv:1709.10259, 2017.
- [7] Ze Xu. Motivic multiplicativity of complete intersections. arXiv:2511.01362v2, revised 30 July 2026, 2026.